Efficient Digital Semantic Communication for Underwater Images Guided by Quantized Rate–Distortion

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Abstract

Underwater acoustic links are narrowband and low-SNR. Digital image transmission therefore usually relies on forward error correction. A channel code has a threshold. Below it, the image is not gradually blurred but lost entirely. We present AdaRoute-VQ, a fully digital image codec that uses no forward error correction and builds error resilience into the bit stream itself. A shared backbone produces a feature map, which two vector-quantization branches represent at different spatial resolutions. The low-resolution branch quantizes with a 256-entry codebook; the high-resolution branch uses a 2048-entry codebook. The feature map is divided into 8x8, that is, 64 routing blocks. Each block selects one branch for quantization. A low-resolution block costs 32 bit. A high-resolution block costs 176 bit. The choice uses the ratio of the two branches' measured quantization errors. A high ratio means the low-resolution branch cannot substitute for the high-resolution one, so the block takes the high-resolution branch. Payload spans a wide range (0.14–0.71 bpp); a target payload maps to a threshold via the formula in Sec. 2.

Every image decodes at every SNR, whereas LDPC-coded JPEG/BPG frequently fail at low SNR and decode only part of the test set at 5 dB. At a matched rate, AdaRoute-VQ clearly outperforms both at 5 dB and is best in PSNR and LPIPS from 5 dB upward. Only at 0 dB is Swin JSCC slightly better; from 5 to 20 dB, AdaRoute-VQ beats it in both metrics while sending slightly fewer bits.

Keywords

underwater acoustic communication, semantic communication, vector quantization, discrete index transmission, rate control

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1 Motivation and Problem

Acoustic links provide only a few kilohertz of bandwidth under severe multipath and Doppler spread. A single 128×128 colour frame therefore occupies seconds of channel time, and only task-relevant content [1, 2] and important content should be sent. Learned joint source–channel coding (JSCC) [3, 4] transmits analog coefficients, which are highly vulnerable to underwater channel impairments, and its indiscriminate transmission protects simple and complex content with the same importance. Vector-quantized codecs [5, 6] produce a plain bit stream and tolerate noise better than continuous features [7], but they are normally paired with a channel code.

We use the term semantic in a narrow, operational sense. A block is important when its low-resolution representation cannot substitute for the high-resolution one, that is, when the distortion caused by substitution cannot be offset by the gain from reduced transmission. The encoder therefore does not need a segmentation mask, saliency map, or ROI side information. This criterion differs from ROI-based allocation and from single-resolution VQ, which spends a fixed number of indices on every cell; instead, we use the reconstruction loss of the block's resolution as the criterion. We report PSNR as a reconstruction measure and LPIPS as the evaluation metric.

2 Method

Two resolution branches. A three-layer convolutional backbone produces $F \in \mathbb{R}^{128\times32\times32}$, which two branches quantize at different spatial resolutions. The feature map is divided into $8 \times 8 = 64$ routing blocks, and each block selects one branch. The payload is

$$\bar{R} = \underbrace{5G}_{320} + \underbrace{C_L b_L}_{32}\, n_L + \underbrace{C_H b_H}_{176}\, n_H, \qquad n_L + n_H = G = 64, \quad (1)$$

The formula decomposes the total payload into three parts: (i) a fixed overhead of $5G = 320$ bit (the routing map repeated five times, 64 bit each); (ii) low-resolution blocks at $C_L b_L = 32$ bit each, n_L blocks; (iii) high-resolution blocks at $C_H b_H = 176$ bit each, n_H blocks. Thus $\bar{R}$ ranges from 2368 to 11584 bit/image. No channel code is used, so payload equals transmitted bits.

Routing and rate control. For block v, let $\bar{e}_{L,v}$ and $\bar{e}_{H,v}$ be the mean squared quantization errors of the two branches. Moving a block from the low-resolution to the high-resolution branch costs $\Delta R = 144$ bit and reduces distortion by $\Delta D_v = \bar{e}_{L,v} - \bar{e}_{H,v}$. The Lagrangian test is therefore the difference rule $\Delta D_v > \lambda \Delta R$. That rule cannot be calibrated to a target rate, because its threshold scales with absolute error. We use instead a scale-free surrogate: a

Table 1: PSNR/LPIPS comparison of four schemes on BCH1/SUIM across SNRs

	Ours PSNR/LPIPS	Swin PSNR/LPIPS	JPEG+LDPC PSNR/LPIPS	BPG+LDPC PSNR/LPIPS
0 dB	17.14/0.393	**17.87**/**0.373**	11.75/0.886	6.65/0.913
5 dB	**22.32**/**0.247**	21.09/0.283	15.87/0.613	14.16/0.623
10 dB	**25.91**/**0.139**	22.72/0.249	23.75/0.206	23.61/0.203
15 dB	**26.91**/**0.119**	23.01/0.249	24.05/0.206	24.71/0.203
20 dB	**27.45**/**0.101**	23.99/0.187	25.44/0.134	24.74/0.138

block takes the high-resolution branch when

$$r_v := \bar{e}_{L,v}/\bar{e}_{H,v} > \tau. \tag{2}$$

For block v, the formula computes the ratio r_v of the mean squared quantization errors of the low-resolution and high-resolution branches. When this ratio exceeds the threshold τ, it indicates that the error of the low-resolution branch is much larger than that of the high-resolution branch (i.e., the low-resolution branch cannot substitute for the high-resolution one), so the block selects the high-resolution branch. Its value lies in invertibility, from which we can derive the following mapping. With $p(\tau) = \Pr[r_v > \tau]$ the payload is $\bar{R} = 2368 + 9216\,p$. A target $\bar{R}^*$ therefore fixes $p^* = (\bar{R}^* - 2368)/9216$ and

$$\tau^* = F_r^{-1}(1 - p^*), \tag{3}$$

where F_r is the empirical CDF of r_v, fitted once on the training split. In this way, we can compute the threshold from a given target rate via the above formulas, and further, a single frozen network covers the entire rate range (0.14–0.71 bpp), because only τ needs to be changed, without retraining.

Protection. We use BPSK, block-interleave the payload over the OFDM resource, and repeat the 64-bit routing map five times with majority voting.

3 Results

Protocol. We use the full 110-image SUIM test set [9] at 128×128; the physical layer is BPSK/OFDM with 512 subcarriers, a cyclic prefix of 256, and one pilot in eight symbols. Metrics are PSNR and LPIPS [11].

Rate accounting. All rates count bits at the physical-layer input and include all protection. All schemes send the same number of bits at the physical-layer input. Ours spends only a small fraction on the repeated routing map, whereas JPEG/BPG spend a quarter on LDPC parity, leaving less for the source. All are matched at 0.496 bpp.

Decodability. JPEG and BPG decode none at 0 dB, 38% and 57% of images at 5 dB, and 92% and 95% at 10 dB.

Learned baseline. Our payload is 8128 bit against 8192 bit for Swin JSCC [10], so we carry about 0.8% fewer bits. Despite this, PSNR is higher by 1.2 to 3.9 dB from 5 to 20 dB, and LPIPS is lower by 0.04 to 0.13.

Different SNR. At 5 dB and matched rate, our PSNR exceeds JPEG and BPG by a wide margin, and LPIPS is the lowest of the four schemes from 5 dB upward. At 10 and 20 dB the margins over both baselines narrow but remain positive, and LPIPS stays lowest. Exact values are in Table 1.

Rate control. One frozen network follows a monotone frontier: at 20 dB on BCH1, the 8128-bit operating point reaches 26.95 dB and the 11584-bit upper end extends the frontier further, spanning 0.14 to 0.71 bpp without retraining. At the matched rate the frontier stays above both intra baselines at 20 dB. The sweep therefore provides rate control and decodability while preserving the frontier advantage.

Ablations. Removing interleaving degrades low-SNR PSNR noticeably. Cutting routing-map repetition hurts more, and severely so when repetition is removed entirely. Replacing error-driven with mask-driven routing also loses PSNR at the same rate. These results support the block-interleaved, repeated routing map and the error-driven routing rule.

4 Conclusion

We presented AdaRoute-VQ, a fully digital image codec for underwater acoustic links. Its core idea is to make error resilience a property of the bit stream itself: the payload is sent as VQ indices, while a short routing map is repeated and majority-voted so that the decoder always knows how to reconstruct the image. The routing map is driven by the ratio of the two branches' quantization errors, which yields a mapping from a target payload to a single threshold and can be deployed without retraining. The resulting image quality degrades gradually as the SNR decreases rather than collapsing abruptly. These properties make AdaRoute-VQ a practical building block for rate-controlled semantic image transmission over narrowband underwater acoustic links.

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